DIABI MOHAMED ZAKARIA

Full-Stack, Mobile & AI Developer

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PROFILE

Computer Science student at Università del Piemonte Orientale with solid hands-on experience in full-stack, mobile, and distributed software development. I design and build real applications spanning backend systems and cloud infrastructure to mobile apps and artificial intelligence, with a particular interest in generative models, LLM-based pipelines, and semantic search.

EDUCATION

Università del Piemonte Orientale — Vercelli — 2023 — ongoing
BSc in Computer Science

Liceo Scientifico — 2022
Scientific High School Diploma

EXPERIENCE

eFarm Group SRL, Milan — Curricular Internship, AI Search Module — Jul 2026 — Sep 2026 (3 months)
Built the AI-powered search module for FotoPortale.it, a platform connecting brands and agencies with creators. Developed the full pipeline: profile data sync, context-text generation, embeddings and vector search combined with structured filters, plus the engine that generates a natural-language explanation of each result via an LLM. Wrote unit and integration tests with PHPUnit in a Dockerized PostgreSQL/pgvector environment.
Technologies: PHP, Symfony, Twig, JavaScript, LLM API, PostgreSQL, pgvector, Doctrine, Docker, PHPUnit

PROJECTS

MiCasaEsTuCasa — 2026
Airbnb-style Android app for managing accommodations: search, booking, and listing management. Client-server architecture with Firebase authentication and real-time data.
Technologies: Kotlin, Firebase, Android

DartHub — 2025 — 2026
Platform for managing darts matches and tournaments, connecting physical games to the internet via IoT sensors. 7-microservice Java architecture with an API Gateway, real-time MQTT communication, and an edge component with offline mode. University lab project (PISSIR).
Technologies: Java, Javalin, JWT, SQLite/JDBC, MQTT, API Gateway, Microservices, Alpine.js

Compilator AC/DC — 2026
Compiler that translates the AC language into DC, implementing lexical analysis, parsing, and code translation phases. Advanced academic project on programming-language architecture.
Technologies: Java, Compilers, Parsing | github.com/Diabi-Mohamed-Zakaria/Compilator-AC-DC

StudyMind — 2026
Multi-agent system for personalized study sessions: generates synthetic notes and calibrated questions from uploaded material, and answers using the Socratic method instead of giving direct solutions. Dynamically adapts to detected gaps and produces a final performance report.
Technologies: Python, Agno, LLM, AI Agents, Pipeline

EventHub — 2025 — 2026
Distributed event-management system built in Java with an RMI architecture, with distinct roles for organizers and attendees. Software Engineering project built with Agile methodology and UML documentation.
Technologies: Java, RMI, UML, Agile | github.com/Diabi-Mohamed-Zakaria/EventHub-Software-Engineering

OggettIUtilInVendita — 2026
iOS marketplace app for buying and selling second-hand items, similar to Subito.it. Listing management, category search, user profiles, and buyer–seller messaging.

Technologies: Swift, iOS, Firebase

Iron Gym Website — 2025
Full web application for managing a gym, with dedicated panels for administrators, trainers, and members. HTML5/CSS3 frontend with a Node.js backend exposing a REST API.
Technologies: Node.js, JavaScript, HTML5/CSS3, REST API | github.com/Diabi-Mohamed-Zakaria/IRON-GYM-WEBSITE

Database Project — 2024 — 2025
Design and implementation of relational databases with PL/pgSQL: from ER modeling to writing complex procedures and queries in PostgreSQL.
Technologies: PostgreSQL, PL/pgSQL, SQL | github.com/Diabi-Mohamed-Zakaria/PROGETTO-DATA-BASE

ProgettoParadigmi — 2024
Java digital bulletin-board system for posting, searching, and managing buy/sell listings, with equivalent CLI and GUI (Swing) interfaces. Data persistence via serialization, a JUnit 5 test suite, and an architecture built on the Strategy and Template Method patterns. Group project.
Technologies: Java, Swing, JUnit 5, Serializable, Design Patterns | github.com/Diabi-Mohamed-Zakaria/PROGETTO-JAVA

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, PHP, Kotlin, C, Swift, SQL

Web & Backend: Node.js, Express.js, REST API, HTML5/CSS3, EJS, Agno, PHP, Symfony, Javalin, MQTT (Mosquitto)

Cloud & Database: Firebase, Google Cloud, PostgreSQL, pgvector, Firestore, SQLite/JDBC, Docker

Mobile: Android (Kotlin), iOS (Swift), SwiftUI

Frontend: Vanilla JS, Alpine.js

Tools & Methods: Git/GitHub, Agile/Scrum, UML, Linux, PHPUnit, JWT, API Gateway, Microservices

AI & Automation: Prompt Engineering, LLM/Chatbot AI, Machine Learning, Vector Search, Embedding

LANGUAGES

French (Native) | Arabic (Native) | Italian (Fluent) | English (B2)